Noise Covariance as Representation Geometry: One Metric, Four Architecture Classes

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Abstract

Under additive Gaussian noise and constrained low-rank reconstruction, measured observations can supply their own reconstruction supervision: the instrument-characterized noise covariance then determines which residuals are informative. For Gaussian noise, inverse covariance is the likelihood-consistent metric. Rather than proposing a new estimator, we organize classical methods as a pair $(\Sigma^{-1}, \mathcal{F})$ that separates measurement geometry from admissible model capacity, and follow one metric across four classes: optimal filtering, unrestricted linear projection, its Kronecker-separable restriction, and factored attention. The linear classes admit an exact geometric characterization. At the Kronecker-separable class, models sharing an architecture but differing only in the training metric reverse their ranking under raw MSE and the likelihood residual, in a frozen ten-seed experiment and a disjoint twenty-seed replication cohort, and under a covariance estimated from $2{,}048$ noise-only events. Each model leading under its own objective is the weaker claim; the substantive result is truth-referenced: the covariance-weighted factors recover the *planted* channel subspace almost exactly and show better exploratory recovery of planted signal parameters, while temporal recovery stays weak. At the attention class the reversal does not appear — a two-seed pilot favours the covariance-trained arm under *both* metrics, and the benchmark score cannot separate the arms — so this bounds the claim rather than extending it.

1 Introduction

Many scientific instruments produce abundant measurements but no labels or clean targets: each observation is its own training signal, and an instrument-derived noise model says which discrepancies matter. Uncertainty varies across frequencies, sensors and conditions; mean-squared error (MSE) weighs all residual coordinates equally, yet under colored noise a large residual can lie in a poorly measured direction while a small one is decisive. The measurement covariance is therefore a measurement-model prior fixing the geometry of reconstruction error, distinct from a conservation law or equivariance [Bronstein et al., 2021, Karniadakis et al., 2021].

This complements self-supervised reconstruction without clean targets [Batson and Royer, 2019, Yaman et al., 2020, Gruber et al., 2025], which learn a denoiser rather than vary the residual metric at fixed capacity, and so are context here, not comparators. Augmentation- and masking-based objectives [Chen et al., 2020, He et al., 2022] construct supervision by transforming an observation; here the measurement supplies the target and an independently calibrated noise model supplies the residual weighting. We ask a controlled question: if an instrument-derived covariance fixes the training metric, how does its effect depend on the admissible representation class?

Every link is classical, and we state the boundary plainly. Inverse-covariance weighting is generalized least squares [Aitken, 1936] and, at rank one, the matched filter [Turin, 1960, Kay, 1993, Szymkowiak

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et al., 1993] — which fixes the weighting but assumes a *known*, one-dimensional template. The latent-variable line relaxes the signal model but not the noise model: probabilistic PCA assumes isotropic noise and factor analysis a diagonal covariance, while weighted low-rank models, Mahalanobis low-rank representation, the linear-autoencoder/PCA correspondence and multilinear PCA are all established [Tipping and Bishop, 1999, Rubin and Thayer, 1982, Roweis, 1998, Bailey, 2012, Allen et al., 2014, Zhang et al., 2019, Baldi and Hornik, 1989, Lu et al., 2008]. *We introduce no new estimator.* Our contribution is a precise separation of measurement geometry from representation capacity, and the finding it makes testable: at matched capacity the covariance-weighted model's held-out ranking reverses — reproduced in a disjoint twenty-seed cohort and surviving finite covariance estimation — and, at the separable condition, it recovers planted structure the isotropic one does not.

The pair $(\Sigma^{-1}, \mathcal{F})$ makes that separation precise: the measurement likelihood fixes the residual geometry Σ^{-1}, the architecture the admissible class $\mathcal{F}$ — including the axis along which the data are tokenized (Section 3). Holding Σ^{-1} fixed, we vary $\mathcal{F}$ across the four classes of Section 3. Among the linear classes the Kronecker-separable family restricts the unrestricted family, and the fixed-template OF case is recovered under Corollary 1's assumptions; all three admit an exact geometric characterization (the separable joint optimization is nonconvex; its subproblems are exact), including when the restriction attains the unrestricted optimum (Proposition 2). The fourth is an analogous axis-factored architecture, examined empirically. Nonlinear classes, distribution shift and adaptive sampling remain future work.

2 Covariance-weighted representation geometry

Let $x_i = s_i + n_i \in \mathbb{R}^d$ be mean-subtracted with $n_i \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$ (mean handling in Appendix B.2). For a candidate $\hat{s}_i$ the Gaussian negative log-likelihood is, up to constants,

$$\chi^2(\hat{s}_i) = (x_i - \hat{s}_i)^\top \Sigma^{-1} (x_i - \hat{s}_i). \tag{1}$$

Whitening, $\tilde{x}_i = \Sigma^{-1/2} x_i$, maps this form to $\|\tilde{x}_i - \hat{\tilde{s}}_i\|_2^2$: Euclidean projection in whitened coordinates, Σ^{-1}-orthogonal projection in measurement coordinates. We report the per-coordinate normalized residual $\bar{\chi}^2 = (Nd)^{-1} \sum_{i=1}^N \chi^2(\hat{s}_i)$, or the per-trace mean where stated.

Why label-free learning works: at the population level, for independent zero-mean signal and noise, $\mathrm{Cov}(\Sigma^{-1/2} x) = \mathrm{Cov}(\Sigma^{-1/2} s) + I$ (Appendix A); adding I shifts eigenvalues, not eigenvectors, so the leading whitened eigenspace of the *noisy* data is the signal's own; at finite N the sample eigenvectors are perturbed. In practice Σ comes from calibration or noise-only records, estimated without evaluation data; a misspecified Σ is a misspecified geometry.

Proposition 1 (Noise-weighted projection bridge)**.** *Under the model above, OF, rank-k covariance-weighted PCA (CW-PCA), and a tied noise-aware linear autoencoder minimize Eq.* (1) *over different admissible subspaces. OF fixes a rank-one template; CW-PCA learns a rank-k subspace; and the tied autoencoder parameterizes the same learned projection when its encoder is the Σ^{-1}-adjoint of its decoder.*

Proof sketch. Whitening reduces the problem to ordinary rank-k approximation, so every minimizer spans a leading eigenspace of the whitened sample covariance; the tied Σ^{-1}-adjoint encoder follows. Projection onto a *fixed* rank-one template yields the OF amplitude; rank-one CW-PCA instead *learns* the leading whitened direction, and the two coincide under the separable-template and eigengap assumptions of Corollary 1 (Appendix A).

Metric reversal. Raw MSE replaces Σ^{-1} by I and is likelihood-correct only when $\Sigma = \sigma^2 I$; under colored noise MSE follows raw variance while Eq. (1) emphasizes high-information directions. Consequently, two equal-rank models can satisfy $\mathrm{MSE}(A) < \mathrm{MSE}(B)$ but $\chi^2(A) > \chi^2(B)$: each is optimal in a different geometry.

3 Four representation classes

Holding the metric fixed, we vary $\mathcal{F}$ across four classes, labelled rungs 1–4. The labels are *not* a capacity ordering: rung 3 *restricts* rung 2, rung 1 is its degenerate fixed-template case, and rung 4 is not comparable. Rungs 1–3 are stated in general two-axis form and evidenced synthetically; rung

4 uses the short-time Fourier transform (STFT) band–frame plane of TIDMAD [Fry et al., 2024], a public release of SQUID readout series from an axion dark-matter search, where injected signals must be recovered from instrument noise and Σ^{-1} is diagonal in frequency by measurement (Section 4).

Rung 1: rank-one fixed template (optimal filtering). Fixing the signal to a known template s_0, the optimal filter (OF) gives the likelihood-optimal amplitude estimate $\hat{A}$ [Gatti and Manfredi, 1986, Turin, 1960], with reconstruction $\hat{s} = \hat{A}s_0$; the representation class is a single direction and the metric question has a closed-form answer.

Rung 2: unrestricted rank-k linear. Replacing the fixed template with a learned rank-k subspace gives covariance-weighted PCA (CW-PCA), which is also the tied noise-aware linear autoencoder: both minimize the likelihood residual over the full Grassmannian $\mathrm{Gr}(k, n_1 n_2)$ of a two-axis measurement (Proposition 1), with reconstruction $\hat{s} = PP^\top \Sigma^{-1} x$ for a Σ^{-1}-orthonormal basis P.

Rung 3: Kronecker-separable two-axis projection (NFPA). For axis-aligned noise, $\Sigma = \Sigma_2 \otimes \Sigma_1$, NFPA restricts rung 2's projection to separable factors of a two-axis measurement $Z_i \in \mathbb{R}^{n_1 \times n_2}$ — sensor channel $\times$ time in the controlled synthetic study, frequency band $\times$ time frame on the public benchmark. On the per-axis whitened observation $\tilde{Z}_i = \Sigma_1^{-1/2} Z_i \Sigma_2^{-1/2}$ it learns orthonormal bases $U_1 \in \mathbb{R}^{n_1 \times k_1}$, $U_2 \in \mathbb{R}^{n_2 \times k_2}$, with code $C_i = U_1^\top \tilde{Z}_i U_2$ and reconstruction $\hat{\tilde{Z}}_i = U_1 U_1^\top \tilde{Z}_i U_2 U_2^\top$ mapped back through $\Sigma^{1/2}$. Figure 2, in the appendix, draws that restriction.

Proposition 2 (Kronecker-restricted bridge)**.** *NFPA minimizes the CW-PCA/tied-autoencoder objective restricted to Kronecker-separable two-axis subspaces. Its optimum equals the unrestricted CW-PCA optimum if and only if at least one rank-$k_1 k_2$ CW-PCA minimizer has such a (k_1, k_2)-separable subspace.*

Proof sketch. In whitened coordinates the factored map is the orthogonal projector onto $\mathrm{range}(U_2) \otimes \mathrm{range}(U_1)$ (vec identity, Appendix A), so NFPA is the whitened rank-$k_1 k_2$ objective restricted to that product; equality holds precisely when that set contains an unrestricted minimizer, and otherwise the gap is the cost of the architecture prior, not a failure of the metric — a cost Appendix B.2 measures by fitting rungs 2 and 3 at matched rank. This is a covariance-aware member of the multilinear-PCA family [Lu et al., 2008], not a new factorization. Even a full-rank factor leaves the restriction active: with $k_1 = n_1$ the family is $\mathrm{range}(U_2) \otimes \mathbb{R}^{n_1}$, still separable. At $k_1 = k_2 = 1$ rung 1 is recovered under Corollary 1's assumptions (Appendix A).

Rung 4: factored attention. Rung 3's restriction on the band–frame plane reads as $\mathrm{range}(U_m) \otimes \mathrm{range}(U_f)$, with U_f over bands and U_m over frames; rung 4's transformer performs intra-band temporal attention over frames, then cross-band attention. The attention model is an *analogous axis-factored architecture* in the same coordinates, and Σ^{-1} is diagonal in exactly the basis U_f indexes: on this benchmark a stationary-PSD model fits the readout noise ($\chi^2/\mathrm{dof} = 1.0031$; Section 4). A token-axis design principle follows: choose the token axis in which Σ^{-1} is diagonal and the likelihood metric becomes a per-token weight rather than a dense coupling, so real and imaginary components are two features of each of 328 physical-frequency tokens, not 656 unrelated rows; this selects a physical token axis rather than proposing a learned discrete vocabulary. The model is a transformer [Vaswani et al., 2017] over frame patches; it is a vehicle, not a contribution (Appendix B.1).

4 Evidence

The controlled experiment. Each event is a $C = 8$ channel $\times$ $T = 256$ sample trace: a pulse with per-event amplitude, rise and decay time, generated around a planted rank-$(k_c, k_t) = (2, 3)$ factorization and corrupted by Brownian $1/f^2$ noise with correlated channels at SNR 10; $800/400/2000$ events train, validate and test; each seed resamples signals, noise and splits. *Two conditions, not to be conflated*: the reversal experiment uses nonseparability $\alpha = 0.1$; truth-referenced recovery is evaluated separately at the exactly separable $\alpha = 0$. Σ is oracle unless stated, and every interval resamples *seeds* at 95% (Appendix B.2).

NFPA under channel compression. We compare NFPA, an isotropically trained factored baseline of the same architecture, CW-PCA and PCA at matched rank 6. NFPA and the isotropic baseline

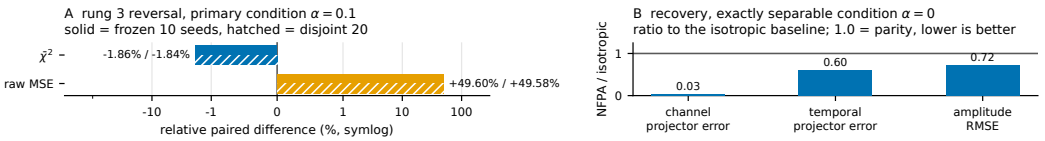


Figure 1: **A**, the primary reversal experiment ($\alpha = 0.1$): relative paired differences, NFPA minus the isotropic baseline; solid = frozen ten seeds, hatched = a disjoint twenty-seed cohort. The metrics point in *opposite* directions — the reversal — and the cohorts agree; symmetric-log axis. **B**, recovery under the *exactly separable* condition ($\alpha = 0$), as NFPA's error over the isotropic baseline's on the same quantity (1.0 is parity): channel almost exact, temporal weak. From archived seed-level results.

differ only in the training metric, and it reverses: in the frozen ten-seed experiment NFPA is better under the likelihood residual (-0.0192, 95% interval $[-0.0194, -0.0190]$) and worse under raw MSE ($+0.0034$, $[+0.0034, +0.0034]$), with narrow seed-bootstrap intervals excluding zero (Appendix B.2). A disjoint twenty-seed cohort, same pipeline and configuration, agrees in sign and magnitude (-0.0190, $[-0.0192, -0.0189]$; $+0.0034$); cohorts are reported separately; a pooled figure would be post-hoc. A twenty-seed white-noise control returns exactly zero, as it must at $\Sigma = \sigma^2 I$. Estimating the covariance from $2{,}048$ noise-only events preserves the reversal in all ten seeds, increasing NFPA's likelihood residual and raw MSE by 0.0078% and 0.0345% respectively of its own oracle values — one fixed condition, not a sweep (Table 2).

Rung 4 (two-seed pilot): the metric in a factored-attention network. Two seeds train one factored-attention architecture on the TIDMAD readout under objectives differing only in the metric. *The reversal does not appear at either seed*: the Σ^{-1}-trained arm is better on *both* criteria (-1.15 and -1.17, both $\times 10^{-4}$, on the likelihood residual; -57.2 and -57.4 on raw MSE, the latter opposite to prediction). The coarse score cannot separate the arms either (Table 1). Of three explanations — degenerate weighting, the architecture, or the instrument's band restriction — only the first is testable here, and measurement rules it out: the per-frequency inverse-noise weights have coefficient of variation 0.69 (range factor 441), the same order as the rung-3 geometry where the reversal *is* observed (1.10; white control 0). Two seeds cannot separate the other two, and both arms sit far below published baselines (Appendix B.1): this bounds the claim, and is not evidence that architecture decides whether the metric matters.

What the covariance-weighted factors recover. The metric also changes what the representation *is*, and these quantities are truth-referenced: unlike the ranking above, none of them follows from each model leading under its own objective. At the exactly separable condition ($\alpha = 0$) NFPA recovers the planted channel subspace almost exactly (0.035 projector error against the isotropic 1.021) while temporal recovery stays weak (0.768 against 1.280) — an asymmetry a joint number would hide. All seven paired contrasts favour NFPA, every across-seed bootstrap interval excludes zero and all ten seeds agree in sign, with no multiplicity correction; two are predeclared (the channel and temporal projector errors) and five post-hoc. No recovery threshold was predeclared either way, so these are continuous estimates, not a verdict (Appendix B.2).

Conclusion. Under additive Gaussian noise inverse covariance is the likelihood-consistent geometry for constrained linear reconstruction. In the controlled setting it changes both the learned subspace and the recovery of planted structure. Separately, the metric reversal persists under one finite-covariance estimate. Whether it survives into competitive nonlinear reconstruction is open — our one pilot did not reproduce it — as is transfer: the same machinery, calibrated synthetically, is over-dispersed on real gravitational-wave strain [LIGO Scientific Collaboration and Virgo Collaboration, 2016, Macleod et al., 2021], so implementation identity is not calibration (Appendix B.3).

Limitations. Rung 4 remains a two-seed pilot (Section 4). The linear-rung TIDMAD reversal is not replicated at a confirmatory file; stationary-PSD adequacy is per-instrument, holding on the TIDMAD readout but failing on GW strain; and one estimated-covariance condition narrows but does not remove the reliance on an oracle Σ — no sweep over calibration sizes or misspecification.

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A Proof details

A.1 Noise-weighted projection bridge

Let $W = \Sigma^{-1/2}P$. The constraint $P^\top \Sigma^{-1} P = I_k$ becomes $W^\top W = I_k$, and the reconstruction satisfies

$$\Sigma^{-1/2}\hat{s} = \Sigma^{-1/2} P P^\top \Sigma^{-1} x = W W^\top \tilde{x}.$$

The empirical loss is therefore $N^{-1}\sum_i \|\tilde{x}_i - WW^\top \tilde{x}_i\|_2^2$. By Rayleigh–Ritz, every minimizer spans a leading rank-k eigenspace of the whitened sample covariance. Mapping back yields the CW-PCA subspace. The objective depends on W only through $WW^\top$, so W is identified up to an orthogonal rotation.

A.2 Whitened signal-plus-identity decomposition

For $x = s + n$ with $s \perp n$, both zero mean, and $\mathrm{Cov}(n) = \Sigma$: independence gives $\mathrm{Cov}(x) = \mathrm{Cov}(s) + \Sigma$, and conjugating by $\Sigma^{-1/2}$,

$$\mathrm{Cov}(\Sigma^{-1/2}x) = \mathrm{Cov}(\Sigma^{-1/2}s) + I.$$

Adding I shifts every eigenvalue by one and changes no eigenvector, so the eigenvectors of the whitened data covariance are those of the whitened *signal* covariance, in the same order. Rank-k CW-PCA on noisy data therefore recovers the leading whitened signal eigenspace at the population level, which is the precise sense in which the likelihood metric supplies a self-supervised training signal.

A.3 Kronecker-restricted bridge

Throughout, the notation is Section 3's: a two-axis measurement $Z \in \mathbb{R}^{n_1 \times n_2}$ with factors $U_1 \in \mathbb{R}^{n_1 \times k_1}$ and $U_2 \in \mathbb{R}^{n_2 \times k_2}$. Section 4's controlled experiment instantiates the first axis as sensor channel and the second as time, so $(n_1, n_2) = (C, T)$ and $(k_1, k_2) = (k_c, k_t)$ there and in Appendix B.2.

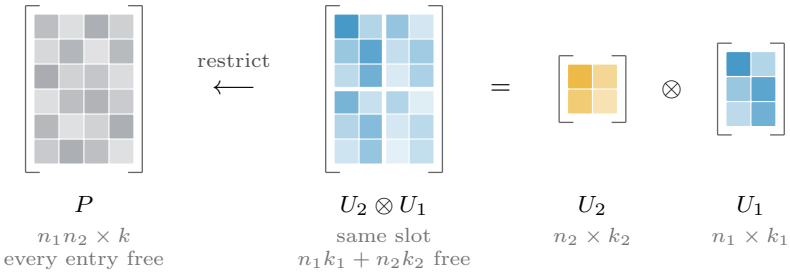


Figure 2: What the restriction is. Both matrices are a rank-k basis, $k = k_1 k_2$, for the same vectorised two-axis measurement, so they occupy one slot; on the right the entries are not free, the basis being a block grid whose every block is a single U_1 motif rescaled by one entry of U_2. That is why the admissible family is the embedded product $\mathrm{Gr}(k_1, n_1) \times \mathrm{Gr}(k_2, n_2) \hookrightarrow \mathrm{Gr}(k_1 k_2, n_1 n_2)$ rather than the full Grassmannian, and why it costs $n_1 k_1 + n_2 k_2$ parameters instead of $n_1 n_2 k_1 k_2$. Cell counts show axis structure, not Section 4's sizes ($n_1 = 8$, $n_2 = 256$, $k_1 = 2$, $k_2 = 3$).

For every such Z, the vectorization identity gives

$$\mathrm{vec}(U_1 U_1^\top Z U_2 U_2^\top) = (U_2 U_2^\top \otimes U_1 U_1^\top)\mathrm{vec}(Z).$$

Column-stacking vectorization puts the *second* axis first, which is why the admissible family is $\mathrm{range}(U_2) \otimes \mathrm{range}(U_1)$ and not the reverse. The Kronecker product is a symmetric idempotent matrix of rank $k_1 k_2$, hence the orthogonal projector onto that subspace. NFPA is thus the ordinary whitened rank-$k_1 k_2$ projection loss restricted to product subspaces. The restricted and unrestricted minima coincide exactly when an unrestricted minimizing projector belongs to this subset.

Table 1: TIDMAD coarse denoising scores (higher is better). **This benchmark's own score cannot separate the two rung-4 arms**, which share architecture, data, and budget and differ only in the training objective: they return bit-identical values because the upstream aggregation rounds the raw score to 0.01 before the logarithm and both band-limited arms fall in the same bin, and rescoring without the rounding gives 0.0241 (`T-mse`) vs. 0.0231 (`T-chi2`) in band, a contrast whose bootstrap interval spans zero (Section 4). Published baselines [Fry et al., 2024] and the archived linear estimate are full-band and so not comparable to the band-limited arms.

Method	Coarse score
FCNet	6.55
Transformer	4.18
PUNet	3.84
none	1.10 (1.1026)
linear Wiener (archived)	-1.8025
`T-mse`	-2.7708
`T-chi2`	-2.7708

A.4 Rank-one correspondence

For $Z_i = A_i v t^\top + \epsilon_i$ with isotropic whitened noise and a unique leading signal eigendirection separated by a nonzero eigengap, the population second moment has leading vectorized direction $(t \otimes v)/(\|t\|\|v\|)$, which is feasible for rank-one NFPA. Its projection coefficient is proportional to $\langle \mathrm{vec}(Z_i), t \otimes v \rangle$, the whitened matched-filter coefficient and hence the OF amplitude up to normalization.

Corollary 1 (Rank-one correspondence)**.** *Assume isotropic whitened noise and a unique leading population signal eigendirection separated by a nonzero eigengap. If that direction is generated by a separable template $\tilde{s} = vt^\top$, population NFPA with $k_1 = k_2 = 1$ learns $v/\|v\|$ and $t/\|t\|$ up to signs; its scalar code is proportional to the OF amplitude of Section 2.*

B Experiment details and reporting limits

B.1 TIDMAD rung-4 baselines

The attention model is a transformer [Vaswani et al., 2017] over frame patches [Nie et al., 2023], with rotary temporal embeddings [Su et al., 2024] and learned band embeddings, trained under two objectives differing only in the metric — `T-mse` on the unwhitened spectrogram, `T-chi2` on the Σ^{-1}-weighted residual — at matched data, budget and evaluation grid.

A full-length empirical study should extend this pilot across additional seeds, splits, noise regimes, ranks and model capacities, reporting intervals for held-out $\bar{\chi}^2$, raw MSE and subspace angles. No such sweep is claimed here.

B.2 Predeclared confirmatory protocol

Mean handling, all experiments: the training mean is estimated once, applied unchanged to held-out observations, and added back after reconstruction.

The multi-seed confirmatory study predeclared three hypotheses. **Executed:** H1 (metric reversal at fixed architecture): under colored or correlated noise, NFPA and an isotropically trained multilinear projection with the same (k_c, k_t) should disagree in ranking, with paired across-seed confidence intervals excluding zero in the predicted directions. H2 (white-noise identity check): when $\Sigma = \sigma^2 I$ the two objectives coincide up to scale, so covariance-aware and Euclidean projectors must agree to numerical tolerance; this validates the pipeline rather than the hypothesis. **Executed after the main blocks:** H3 (active restriction): with $C > k_c$ and $T > k_t$, NFPA should recover the planted channel and temporal subspaces for separable signals. This was not run with the original blocks; it was executed afterwards on the archived fits, each of which reproduces its archived held-out residual to $\leq 10^{-9}$ relative, so the analysis is computed on the archived fits rather than on refits that merely resemble them. Because the protocol set no numerical recovery threshold, the result is reported as continuous projector errors with bootstrap intervals that resample seeds and no binary

verdict. Truth-referenced parameter decoding (amplitude, rise time, decay time) was proposed in a block dropped before execution; it is reported at the archived $\alpha = 0.1$ condition as *post-hoc exploratory*, and at the separable $\alpha = 0$ condition alongside the H3 analysis. An earlier design specified 30 seeds; the executed, frozen protocol is 10 seeds, amended before any confirmatory result existed. The primary configuration uses $C = 8$ channels, $T = 256$ samples, ranks $(k_c, k_t) = (2, 3)$, $800/400/2000$ training/validation/test events, 10 independent seeds, 10 ALS restarts per seed, SNR 10 under $\mathrm{SNR}^2 = \mathbb{E}[s^\top \Sigma^{-1} s]$, Brownian $1/f^2$ temporal noise with channel correlation, oracle covariance, and nonseparability $\alpha = 0.1$. Results enter the main text only if every acceptance gate holds: all seeds complete without post-result configuration changes; the fixed-architecture reversal intervals exclude zero in the predicted directions; the white-noise check passes its predeclared equivalence margin; and the table and figure are generated directly from archived seed-level results.

Standardized effect size at rung 3. The relative differences in Section 4 are small against the *level* of the normalized residual ($\bar{\chi}^2 \approx 1.014$) and large against seed-to-seed variation. With d_z the mean paired difference divided by the standard deviation of the per-seed differences, the frozen cohort gives $d_z = -55.1$ on $\bar{\chi}^2$ and $+99.5$ on raw MSE, the replication cohort -66.2 and $+71.2$, and the estimated-covariance condition -52.6 and $+93.6$. Conventionally $|d_z| = 0.8$ is a large effect. This is also why every cohort has frac(sign agrees) $= 1$: the reversal is essentially deterministic across seeds at this configuration. It is a statement about consistency, not about practical importance at any particular instrument.

Rung 3 against the unrestricted rung-2 optimum. At the frozen $C = 8 > k_c = 2$ configuration NFPA's held-out likelihood residual is lower than unrestricted CW-PCA's by 0.0026 ($[0.0025, 0.0026]$). This held-out finite-sample difference does not imply that a restricted training optimum beats the unrestricted one: Proposition 2 compares the fitted objective over nested classes and gives the condition for equal optima.

Semantic recovery, per method. At the separable condition NFPA attains amplitude RMSE 0.143 against the isotropic baseline's 0.198 (decoded-amplitude correlation 0.70 against 0.13), and lower rise- and decay-time errors. The same analysis at the archived $\alpha = 0.1$ condition is post-hoc exploratory and agrees in direction on all seven contrasts.

The factored representation (Section 3, generic form) has $O(k_1 n_1 + k_2 n_2)$ basis parameters rather than $O(k_1 k_2 n_1 n_2)$ for an unstructured basis, trading approximation flexibility for a two-axis inductive bias related to classical tensor factorizations [Kolda and Bader, 2009, Lu et al., 2008]. Alternating spectral updates solve each factor subproblem exactly, although the joint product-manifold objective remains nonconvex.

B.3 Transfer of the noise model

On a public gravitational-wave strain release [LIGO Scientific Collaboration and Virgo Collaboration, 2016, Macleod et al., 2021], a whitened score matching its reference form to 10^{-12} and calibrated on synthetic stationary noise (dispersion 1.15) is seven to nine times as dispersed on real H1 and L1 strain (9.42 and 8.83; the global-PSD arm gives 1.77 and 7.29). The claim is about covariance adequacy on that instrument, not about NFPA and not about detector sensitivity.

B.4 Estimated covariance, and a corrected angle diagnostic

Table 2: **Estimated covariance** — a single frozen sensitivity condition: the separable covariance is estimated from 2,048 independent noise-only calibration events rather than taken as oracle, at the otherwise unchanged $C = 8 > k_c = 2$ configuration. The reversal survives in all ten seeds. The degradation percentages are relative to NFPA's *own* corresponding archived oracle held-out value. This is one condition, not a sweep across calibration sizes, covariance families or misspecification. **Corrected angles** — a post-hoc diagnostic correction of the whitened-basis vectorization used for principal angles. Archived as `E10b` and `E11` in the run records.

Quantity	Estimate
Estimated separable covariance, $N_{\mathrm{cal}} = 2048$, 10 seeds	
$\Delta\bar{\chi}^2$, NFPA − isotropic	-0.019122 $[-0.019340, -0.018914]$
ΔMSE, NFPA − isotropic	$+0.0034178$ $[+0.0033963, +0.0034393]$
NFPA $\bar{\chi}^2$ degradation vs. oracle	7.90×10^{-5} (0.0078%)
NFPA MSE degradation vs. oracle	3.55×10^{-6} (0.0345%)
Product-scale ratio vs. oracle	0.9991
Channel / temporal shape error	0.0282 / 0.0197
Kronecker relative Frobenius error	0.0365
Corrected principal angles to CW-PCA (degrees)	
Frozen, $n = 10$	NFPA 62.89, isotropic 86.71, difference -23.82
Replication, $n = 20$	NFPA 63.03, isotropic 86.73, difference -23.71

Note: the correction leaves NFPA unchanged and shifts the isotropic baseline by 0.81° in the frozen cohort and 0.73° in the replication cohort, so no archived scientific conclusion depended on the defect. Displayed intervals are 95% bootstrap intervals with seed as the resampling unit. Cohorts are never pooled for confirmatory reading.

The estimated-covariance condition repeats the frozen $C = 8 > k_c = 2$ configuration with one change: the separable covariance is estimated from $N_{\mathrm{cal}} = 2048$ independent noise-only calibration events instead of being supplied as an oracle. Estimation proceeds by sample estimation of each factor, the regularization already used for the oracle path, and a symmetric product-scale calibration $\hat{v} = \sqrt{v_c v_t}$ with each factor divided by $\sqrt{\hat{v}}$; that correction was frozen before any full result existed. The reversal survives in all ten seeds, and the degradation relative to NFPA's own oracle held-out values is small but resolved — consistent with the fit being invariant to a global whitening scale (the estimated product scale is within 0.1% of oracle) while the covariance *shape* is estimated accurately. This is a single condition: it does not bound behaviour across calibration sizes, covariance families or deliberate misspecification, and no such sweep is claimed.

We corrected a vectorization-order error affecting only the isotropic baseline's principal-angle diagnostic. The correction shifts that angle by less than one degree, leaves NFPA's unchanged, and affects neither primary metric nor any conclusion; the corrected values are the ones reported. It is post-hoc and is not a second attention-class seed.

Separate rung-2 benchmark. In a separate 30-seed rank-two benchmark (512 traces of dimension 256 per seed), PCA and CW-PCA disagreed in ranking: the paired held-out weighted-residual difference favoured CW-PCA with a 95% interval excluding zero, raw MSE preferred the opposite reconstruction, and the two whitened subspaces were nearly orthogonal. A ten-seed white-noise control returned a numerically zero angle. On a single TIDMAD file the covariance-weighted reconstruction reduced the amplitude RMSE by roughly a quarter.

B.5 Sanity check: the earlier single-split study

This predates the primary experiment and does *not* compress the channel axis; it is retained as an implementation sanity check, not as evidence. It uses $C = 2$ channels, $T = 256$ samples, 1 kHz sampling, channel weights $(1.0, 0.65)$, amplitude $A \sim \mathcal{N}(1, 0.3^2)$, Brownian $S_n(f) \propto 1/f^2$ noise, and $800/300$ train/test events; NFPA and IsoAE use $(k_c, k_t) = (2, 3)$, CW-PCA and IsoPCA rank six.

Because $C = k_c = 2$, $U_c U_c^\top = I_2$ here: the reconstruction restricts a shared temporal subspace but does not compress channels. Principal angles between NFPA and CW-PCA are 0.0°, 59.3°, 64.6°,

73.8°, 84.8° and 87.8° (mean 61.7°) — the counterexample showing that a full-rank factor does not deactivate the restriction. The IsoAE–CW-PCA angles are 82.1°, 83.7°, 86.0°, 86.6°, 88.2° and 89.4° (mean 86.0°). These are point estimates from one seed and split, computed by a different code path from the multi-seed cohorts, so the corrected angles of Table 2 are not replacements for them. Held-out raw MSE for IsoAE and IsoPCA is 4.58×10^{-9} and 4.33×10^{-9}.